

A Rolls-Royce
solution

designation
circuit-diagram "control-panel" BR4000 ECU9 / BASLER

project number
XZ54934000028 (ZNG000040837)

material number
see table-item

serial number

type-of-system
SUP42

revision	date	name	checked	creator	E56395 Dieter Enfrlich	17.05.2021	customer-description	project-description circuit-diagram "control-pane l" BR4000 ECU9 / BASLER	 A Rolls-Royce solution Rolls-Royce Solutions GmbH	Page Description Cover sheet	drawing-number BR2000-4000ECU9 material number see table-item	order number	EPLAN P8 2.8.3	== +	version Standard	page (&EAA) Full pages	1 from 2	10 88
			norm		F.Jäger M.Prinz	21.10.2021 12.02.2020												

PROJEKT INFORMATION

Wire labelling: No

Protection: IP54

Mains connection: Rated voltage 400V AC
Mains frequency: 50Hz
Wires of cable: 3 Phase / N / PE
Cross section max.: 5x10mm²
Max load: 22A
Max Icu: 25kA

MTU Part-No		Demonation	
X54934400491 (ZING000040645)		Basler 2020HD 5A X59434200299 / 5NE2DNSBR	
X54934400492 (ZING000040646)		Basler 2020HD 1A X59434200300 / 1NE2DNSBR	

Part list number	remark
X54934400425	Basler mit 1A
X54934400429	Basler mit 1A mit IoT
X54934400432	Basler mit 5A
X54934400433	Basler mit 5A mit IoT

INFORMATION TO SWITCHGEAR ASSEMBLY

	generally	auxiliary-drives	Generator	voltage sensing busbar / mains
type-of-electrical-system	TN-S			
rated operating-voltage Ue:		max. 440 VAC	max. 690 VAC	max. 690 VAC
rated voltage Un:		440 VAC	690 VAC	690 VAC
rated insulation voltage Ui:		440 VAC	690 VAC	690 VAC
rated impulse withstand voltage Uimp		4 kV	6 kV	6 kV

3,0 bis 25,0A

rated current In:

max. back-up-fuse:

rated conditional short-circuit current Icc:

rated diversity factor RDF:

rated frequency fn:

overvoltage-category:

pollution-degree:

EMC ambience:

protection-class:

min. ambient-temperature:

max. ambient-temperature:

max. installation-altitude:

auxiliary-voltage DC:

auxiliary-voltage AC:

Client:	# # #
Series no.:	# # #
Type designation:	MIP BR4000
Material no.:	# # #
Type of electricl system:	TN-S
Rated Voltage Un:	415V AC
Rated Frequency fn:	50 Hz
Rated current InA:	# # #
Rated short current ICC:	25 kA
Control Voltage:	24V DC
Protection class:	IP 54
Year of manufacturing:	# # #
Listed standards:	IEC 61439-1 IEC 61439-2 DIN EN 60204-1

0,08 - 0,68

50 Hz

III

3

A

IP 54

0 °C

45 °C

2000 m

24 V

24 V

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ENG	EFS	4	Engine-Control-Unit ECU9 Plug -X1 E-Stop - analog - power	13.10.2021	M.Mayer
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TECHNISCHE DATEN UND KONSTRUKTIONSDetails

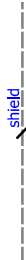
IEC 61355

Vollständige Bezeichnung		Strukturbeschreibung
mounting-location		
+BSF	Grundrahmen	
+ENG	Motor	
+GEN	Generator inkl. Klemmenkasten	
+MIP	Motor-Steuer-Schrank	
+MMC	MTU Modul Steuerschrank	
+GCB	Generator Leistungsschalter	
+MCB	Netz Leistungsschalter	
+FAN	Lüfter Steuerschrank	
+TNK	Kraftstoff-Tank	
+EXT	Nicht im Schaltschrank eingebaute Geräte und Gehäuse die nicht von MTU geliefert werden	
+CUS	Kundenseite	
+DOK	Dokumente	

Vollständige Bezeichnung		Strukturbeschreibung
document-type		
&EAA	Deckblatt	
&EAB	Inhaltsverzeichnis	
&EDB	Beschreibung	
&EFS	Schaltplan	
&ETL	Layout	
&EMA	Klemmenplan	
&EMB	Kabelplan	
&EPC	Materialstückliste	

Vollständige Bezeichnung		Strukturbeschreibung
system		
=EKW	Elektrische Stromversorgung	
=CON	Steuerspannung	
=EES	Maschinen Not-Halt	
=PLC	SPS oder Genset-Kontroller	
=COM	Kommunikation	
=STR	Motor Start	
=MCC	Genisch Kühl-Kreis	
=ECC	Motor/Kühlwasser System	
=HWC	Heiz-Wasser-Kreis	
=FUL	Kraftstoff Versorgung	

TECHNICAL DATA AND CONSTRUCTION DETAILS

line and color definition for potential tracing			(Colors are only visible if you use colored print out)		
Supply, Line and measure voltage	L1, L2, L3 (120V, 230V, 400V ,690V, ...)				
Supply, Line and measure voltage	N (120V, 230V, 400V ,690V, ...)				
Protective earth	Ground				
Control voltage + 24VDC, 12VDC L+	L+				
Control voltage - 0VDC L-	L-				
Shield	SH				
Starter battery cabeling +24V	L-				
Starter battery cabeling -24V	L-				

line and color definition for potential tracing		(Colors are only visible if you use colored print out)	
Control voltage 24VAC 24VAC L		24VAC-L	
Control voltage 24VAC 24VAC N		24VAC-N	
External source voltage		ext	
Analog signals		analog	
Current transformer		L1-k-Geno	
Emergency stop		NA	
All other connections EPLAN standard			

TECHNICAL DATA AND CONSTRUCTION DETAILS

Wire colors within enclosure			
According IEC EN 60204-1			
Rated voltage	Above 48VAC	black	
Neutral wire	N	light blue	
Protective earth	PE	green/yellow	
Control voltage	24-240V AC	red	
	0V AC	red/white	
	24-110V DC	dark blue	
	0V DC	grey	
PLC-Controller		white	
Measuring technique		grey	
wires before Mainswitch		orange	

		L1 - L2 - L3 1L1 - 1L2 - 1L3 N PE
Main power before main switch		
Main power main switch on		
Neutral wire		
Protective conductor		
Mains power supply	400VAC	-XD00
Power supply Heating GEN	230VAC	-XD01
Power supply heatg. cool. water	400VAC	-XD02
Emergency stop		-XD06
Supply DC	24VDC	-XDL+
Supply DC	0VDC	-XDL-
Harness W011-ECU9 Digital		-1XD01
Harness W011-ECU9 Analog		-1XD02
Harness W020 Engine		-2XD01
Harness W030 Generator signal		-3XD01/-3XD02
Harness W031 Generator PT100		-3XD03
Option		
Customer interface WD04.1 / WG04.2 / WG04.3		-4XD01/-4XD02/ -4XD03
Harness W050 Power panel		-5XD01
Control cable WD06.1 Generator circuit breaker		-6XD01
Control cable WD07.1 Mains circuit breaker		-7XD01
Control cable WD08.1 Fan control enclosure		-8XD01
Control cable WD09.1/WD09.2 Fuel control		-9XD01/-9XD02
CAN bus		-XF01

Tightening torque			
Power relay	75A	M3,5x5 M5x8	1,1-1,2 Nm 3,2-3,5 Nm
Starter	Prestolite 9KW	M5 M12	2,0-3,0 Nm 27-33 Nm
Starter	Prestolite 15KW	M5 M10	1,5-1,7 Nm 9-10 Nm
Starter	Delco 9KW	Housing 10-32 INCH 1/2-13 INCH	18-23 Nm 1,8-3,4 Nm 27-34 Nm
Battery disconnect switches		M10 M12	15-20 Nm 18-22 Nm
Battery terminals	DIN 72331	Cable M6 M8	5-6 Nm Class 4,6 / 3,5 Nm Class 4,6 / 8,4 Nm
Grounding		M8 M10 M12 M14	Class 8,8 / 28 Nm Class 8,8 / 54 Nm Class 8,8 / 93 Nm Class 8,8 / 148 Nm